Soccer-Twos Final Project

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Abstract: We compare two PPO-based approaches to the sparse-reward problem in the SoccerTwos 2-vs-2 environment. The primary agent uses *curriculum-guided self-play*, shaping the state distribution by progressively widening ball position, velocity, and opponent difficulty, with stage advancement gated on a performance threshold and a snapshot pool of recent policies. The additional agent uses *seeded historical self-play with potential-based reward shaping*, shaping the reward signal via an exponential ball-to-goal potential while sampling opponents from a mixture of ranked seed checkpoints, run-time history, and the current policy. The curriculum agent and the seeded history agent reaches 86% and 77.5% win rates respectively against the CEIA baseline over 50 games and the curriculum agent reaches 100% against the TA agent. In a direct 50-game head-to-head of our two agents, the curriculum agent wins 72% of games, suggesting that controlling the initial state distribution is more effective than reward shaping alone for this task. Github Code Link

Keywords: PPO, Selfplay, Curriculum Learning

1 Overview

SoccerTwos is a sparse-reward, partially observable, 2-vs-2 competitive environment in which goals are rare events and the opponent distribution is non-stationary under self-play. We frame this as two coupled difficulties: *a sparse exploration signal and non-stationary opponents*. We explore two complementary ways: shaping the *state distribution* the agent sees (curriculum learning) and shaping the *reward signal* it receives (potential-based reward shaping). We evaluated both levers under matched evaluation protocols. **Agent 1** (curriculum-guided self-play PPO) targets the policy-performance criterion. **Agent 2** (seeded historical self-play PPO with potential-based reward shaping) targets the reward-modification criterion. Both agents use PPO with Ray RLlib and a $[256, 256]$ fully connected policy, and both are evaluated against the Random, CEIA, and TA agents, as well as head-to-head against each other.

2 Methods

Both submitted agents share the same backbone — Proximal Policy Optimization (PPO) [1] implemented with Ray RLlib, a $[256, 256]$ fully connected policy, and Generalized Advantage Estimation [2] with $\gamma = 0.99$ and $\lambda = 0.95$ — and differ only in the lever they pull to address the sparse-reward, non-stationary self-play problem. PPO is an on-policy policy-gradient method that stabilizes training via a clipped surrogate objective, which limits destructive updates between consecutive iterations. The primary agent (Sec. 2.1) reshapes the *state distribution* via curriculum learning over environment initializations, gated by performance thresholds, with self-play introduced once basic scoring is acquired. The additional agent (Sec. 2.2) reshapes the *reward signal* via a potential-based ball-to-goal bonus, while drawing opponents from a seeded mixture of ranked checkpoints and run-time history. Presenting the two as complementary lets us isolate which lever — state distribution or reward signal — is more effective when both build on the same PPO backbone.

2.1 Primary Agent: Curriculum-Guided Self-Play PPO

The primary agent's modification is curriculum learning through environment initialization [3]. Early stages constrain the ball to be near the opponent goal with minimal velocity, and later stages progressively introduce wider spatial distributions, increased ball velocity, and more complex player configurations. Crucially, advancement between stages is gated by a performance threshold, so the policy is not exposed to a harder distribution until it demonstrates stable behavior on the current one.

Hypothesis: Gating curriculum advancement on performance prevents premature exposure to difficult opponents, allowing the policy to first acquire reliable goal-scoring behavior under simplified state distributions before generalizing to the full task. This should yield faster convergence and better generalization than training directly against strong opponents.

In the later curriculum stages, the opponent is sampled through self-play. The trainer periodically saves policy snapshots, retains a pool of the most recent 8, and at each episode samples either the current policy (probability 0.2) or one of the historical snapshots (probability 0.8). Holding most of the sampling weight on past snapshots stabilizes training by preventing the agent from continually chasing its own evolving behavior [4].

Learning rate	1×10^{-4}	Selfplay Current Policy Prob	0.2
Batch size	8000	Clip parameter	0.2
Minibatch size	1024	Entropy coeff.	0.003
SGD iterations	6	Value loss coeff.	1.0
Reward Window	25	Grad clip	0.5
Stop Timesteps Total	10000000	γ	0.99
Selfplay Pool Size	8	λ	0.95
Selfplay Snapshot Interval	50		

Table 1: Training hyperparameters for primary agent

The hyperparameters shown in Table 1 were selected to balance training stability, performance, and sample efficiency.

2.2 Additional Agent: Seeded Historical Self-Play PPO

The additional agent uses seeded historical self-play with potential-based reward shaping. The setup uses two RLlib policies, i.e. a trainable `current_team` optimized by PPO and a frozen `opponent_team` whose weights are periodically replaced from a checkpoint pool. The agent side is randomized each episode so the policy learns both blue- and orange-side play.

Hypothesis: Seeding the opponent pool with strong external checkpoints, rather than relying on history-from-scratch, removes the cold-start phase in which a fresh policy plays only weak versions of itself; combined with a dense, goal-directed shaping reward, this should produce stronger CEIA transfer than naive or historical self-play alone.

The opponent pool combines four sources: high-performing seed checkpoints from previous self-play experiments, historical checkpoints from the current run, the current policy itself, and uniformly sampled seed checkpoints. Seed checkpoints are ranked before training, so the trainable policy is exposed to a difficulty distribution that is non-trivial from the first iteration but still varies over time.

To address sparse scoring rewards, we apply potential-based reward shaping with an exponential two-dimensional ball-to-goal potential. For a team whose opponent goal center is g, let b_t be the ball position, $d_t = \|b_t - g\|_2$, $\phi_t = \exp(-d_t/6.0)$, and $\Delta\phi_t = \phi_t - \phi_{t-1}$. The shaped reward is

$$r'_t = r_t + \alpha_1 \max(\Delta\phi_t, 0) + \alpha_2 \min(\Delta\phi_t, 0).$$

The progress term is positive only when the ball moves closer to the opponent goal, the retreat term is negative only when it moves away, and the asymmetric weights ($\alpha_2 > \alpha_1$) penalize retreat more strongly than they reward progress.

3 Experimental Results

3.1 Primary Agent Results

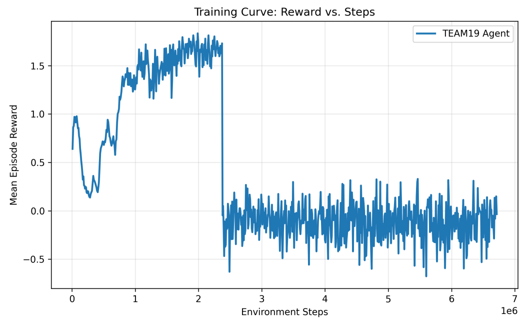


Figure 1: Training curve showing mean episode reward vs. environment steps for Primary agent.

The training curve of primary agent is shown in Figure 1. Training begins with unstable performance. In `close_finish_static`, mean reward rises from 0.64 to 0.97 while episode length decreases, but rewards are bimodal, indicating inconsistent success. As the curriculum progresses, performance drops when entering harder stages, where reward falls and episode length increases, showing adaptation to new dynamics. After adaptation, performance improves significantly in random-opponent stages, with mean rewards exceeding peaking at 1.83 and episode lengths dropping below 100 steps, indicating efficient scoring behavior. However, in `hard_self_play`, performance becomes unstable, with mean reward fluctuating around zero and a bimodal distribution of strong wins and losses. Overall, while the agent learns effective scoring strategies, it remains inconsistent against stronger self-play opponents.

After 779 PPO iterations, 6.71 million environment timesteps, and 82,457 episodes, the agent had reached the hardest curriculum stage, `hard_self_play`, with a self-play opponent pool of size 8. Although mean reward remained unstable in this final stage, the checkpoint still produced successful high-reward episodes, with a maximum reward of about 1.98.

The final agent showed strong performance across evaluation settings. Against the CEIA baseline, it achieved 100%/82%/81% win rate over 10/50/100 games. Against both the random agent and TA agent, it achieved 100% win rates. The policy performed well from both sides of the field, although the evaluation still showed a slight side bias. In the direct 50-game comparison between our two agents, the primary curriculum self-play agent defeated the additional seeded historical self-play agent with a 72% win rate.

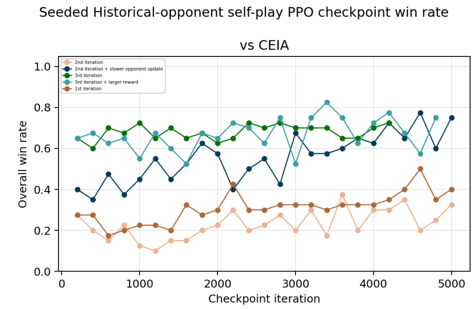
3.2 Seeded Historical Self-Play Results

We trained the agent for several seeded iteration. In each seeded iteration, initial opponents are picked by previous highest-performing checkpoints and the current policy will be initialized with the best checkpoint. The reward curve and win rate curve against CEIA agent is shown in Figure 2, and the progressively improved training method is shown in Table 3 in the appendix. The final checkpoint achieved 80%/77.5%/71% win rates against the CEIA baseline over 10/50/100 games. This is substantially higher than historical self-play-from-scratch (no initialized weight) result of 37.5% and simple self-play result (no history) of 30% against CEIA on 50 games, but it still underperformed our curriculum-guided self-play PPO agent.

3.3 Cross-Agent Comparison

Table 2 summarizes our agents' win rates against the Random, CEIA, and TA agents, as well as the head-to-head result between the two agents. The primary agent reaches the highest reported win rate

(a) Mean episode reward versus environment steps for the selected seeded historical self-play training run.



(b) CEIA win rate versus checkpoint iteration for seeded historical self-play variants.

Figure 2: In the left figure, the reward curve fluctuates around 0 because the training sometimes samples current policy as the opponent.

against every opponent for which both agents were evaluated, and it wins the direct head-to-head with a 72% win rate over 50 games. While both agents substantially exceed their respective naive baselines (cf. Table 3 for the additional agent's development trajectory), this cross-agent comparison indicates that curriculum-guided control of the state distribution yields a stronger final policy in this environment than seeded historical self-play with potential-based reward shaping.

Agent	vs. Random	vs. CEIA	vs. TA	Head-to-head
Agent 1: Curriculum self-play	100%	86%	100%	72%
Agent 2: Seeded historical self-play	100%	77.5%	TBD	28%

Table 2: Evaluation summary for the two submitted agents. Comparisons are conducted with 50 games except for TA agent (10 games). Head-to-head shows comparison between our two agents.

4 Analysis and Discussion

4.1 Effects of combining curriculum learning and self-play

The curriculum and self-play modifications improved training progression by allowing the agent to first learn simple goal-scoring behavior before being exposed to harder opponents. The curriculum reduced the difficulty of early exploration, helping the policy discover useful trajectories in sparse reward conditions. As training advanced, the agent progressed through all curriculum stages and eventually reached `hard_self_play`, indicating that the advancement mechanism successfully exposed the policy to increasingly challenging scenarios. Self-play further increased the difficulty of the training distribution by training the agent against policies sampled from previous checkpoints. This prevented the final policy from being trained only against fixed opponents and encouraged adaptation to a wider range of behaviors. The strong performance against both CEIA and TA agents suggests that the combined curriculum and self-play setup produced a policy that was more effective.

4.2 Analysis of Seeded Historical Self-Play

The seeded historical self-play agent improved over earlier random-opponent, naive self-play, and historical self-play baselines, mainly because it began from a stronger checkpoint and sampled from stronger historical opponents. The two-dimensional shaping reward also aligned dense reward with the true scoring objective better than a one-dimensional progress reward. However, this agent did not outperform the primary curriculum self-play agent as shown in Table 2. This suggests that curriculum control over the initial state distribution solved the sparse scoring problem more effectively than seeded historical self-play and reward shaping alone.

References

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Appendix

Training method	vs. Random	vs. CEIA	Notes
against random team agent	87.5%	17.5%	Fixed weak-opponent baseline
+ reward shaping	90%	25%	still weak transfer
naive self-play + reward shaping	90%	30%	
Historical self-play from scratch	100%	37.5%	weak transfer
1st-iteration seeded agent	100%	50.0%	Seeded opponents improved early transfer
2nd-iteration seeded agent	100%	68%	initialized with best checkpoint from 1st iteration
3rd-iteration seeded agent	100%	77.5%	Best checkpoint against CEIA

Table 3: Evaluation summary for the additional seeded historical self-play agent and its development baselines, all implemented with PPO as the backbone. Evaluations are conducted with 50 games.